Responsibly Deploy AI in Quantitative Social Research

As part of the Social Research Methods module

Rongxin Ouyang*

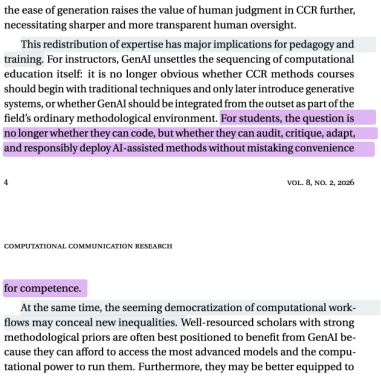
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Using causal and computational methods to evaluate
societal and political impacts of digital information technologies

November 5, 2026; PKU

The goal of this session

(... lower barrier (of computational skills) ... redistribution of expertise ...)

"For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience" (Zhou et al., 2026)



the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

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COMPUTATIONAL COMMUNICATION RESEARCH

for competence.

At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

Source: Zhou et al. (2026)

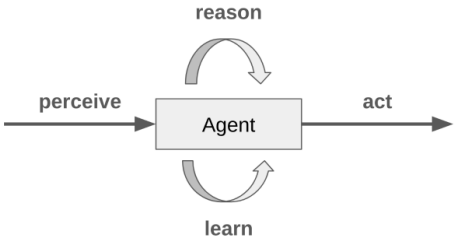
Introduction

1.1 What is artificial intelligence?

- Computational agents that act intelligently

  (Poole & Mackworth, 2023)

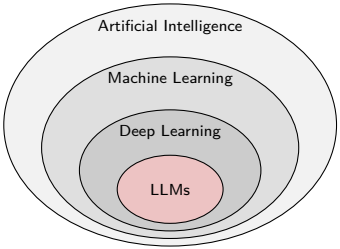
- But what is intelligence? See the figure



Source: Stanford CS221, Percy Liang

1.1 What is artificial intelligence?

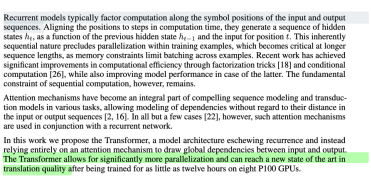
- Computational agents that act intelligently (Poole & Mackworth, 2023);
- But what is intelligence? See the figure
- More importantly, please be aware of the differences ...



1.2 A simplified history of LLMs

- * Artificial intelligence is an old field developed decades ago (see Poole & Mackworth, 2023);
- * Pre-LLMs: early tools mainly classified, clustered, or predicted from human-produced traces (Zhou et al., 2026). [a]
- * Birth of LLMs: the Transformer (Vaswani et al., 2017), a model architecture that parallelizes training, enabling large datasets and model sizes
- Post-LLMs: we focus on the LLM era after 2020 (marked by GPT-3; Xu et al., 2026) and its implications for computational social science research, with a brief tutorial on text classification using RoBERTa

[a] Likewise, Percy Liang uses 2016, 2018, and today to divide three time periods; see https://cs336.stanford.edu/lectures/?trace=lecture_01



Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

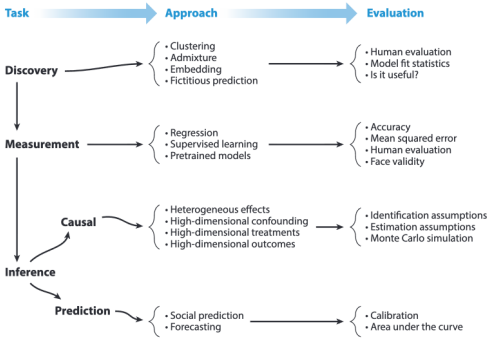
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

Excerpt from Vaswani et al. (2017)

Pre-LLMs AI

2.1 The use of AI in social science before LLMs

- Social science research often leverages machine learning for various tasks, according to the typology in Grimmer et al. (2021):
  - Discovery; e.g., text classification (Tutorial I)
  - Measurement; e.g., regression
  - Causal Inference; e.g., heterogeneous treatment effects
  - Prediction; e.g., forecasting social trends
- Take FIN528 Machine Learning for Finance / Advanced Machine Learning at PHBS if interested



Source: Grimmer et al. (2021) (p.23)

*TUT1

Sentiment classification using RoBERTa

Online material:
Google Colab Link

LLMs

3.1 Roles of LLMs in Computational Social Science

- Large language models (LLMs) are language models with roughly 1 billion or more parameters (Percy Liang; e.g., GPT, Claude)[1]
- A recent review article suggests four roles of LLMs in computational social science: subject, assistant, collaborator, and evaluator

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

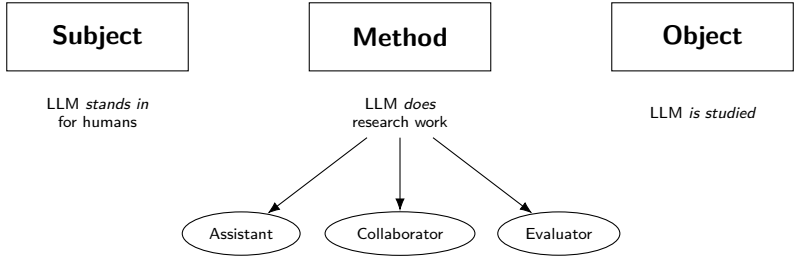
Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

Source: Xu et al. (2026)

[1] This is not a CS class, so it should be fine to use this definition.

3.1 Roles of LLMs in Computational Social Science

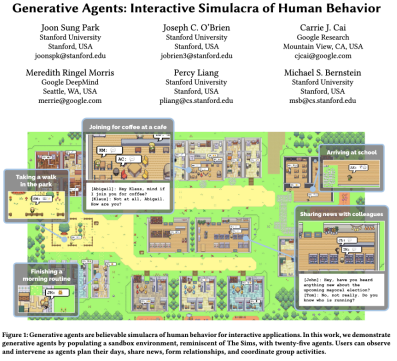
- Drawing on more perspectives, we propose a comprehensive categorization of LLM roles in CSS, elaborated in the following slides



Source: Four roles (subjects, assistants, collaborators, evaluators) categorized in Xu et al. (2026); four categories in Hai Liang (2026); and two roles (a research tool, an object of inquiry) adapted from Zhou et al. (2026)

3.2 LLMs as Subjects: Categories

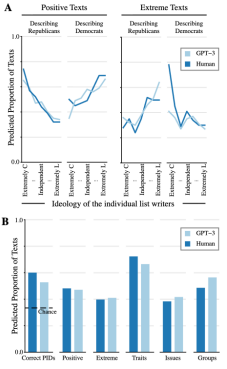
- LLMs serve as research subjects, according to Xu et al. (2026):
  - Generating survey responses (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - Simulating behavior sequences (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - Simulating social interactions (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



Generative agents in a simulated town; Park, O'Brien, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

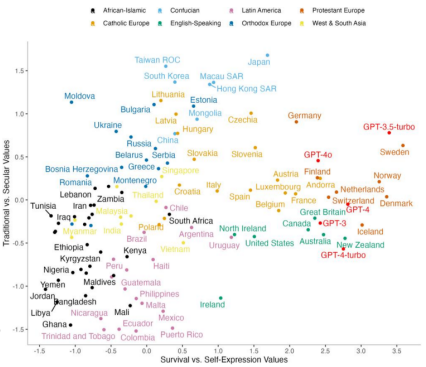
- **Pros:**
  - Successfully replicate human responses in surveys (Argyle, Busby, et al., 2023)
  - Produce believable individual and emergent social behaviors (Park, O'Brien, et al., 2023)
  - Replicate emergence of social conventions (Ashery et al., 2025)
  - etc.



Analysis of GPT-3 and human responses from the Lucid survey; Fig 3 in Argyle, Busby, et al. (2023)

3.3 LLMs as Subjects: Pros and Cons

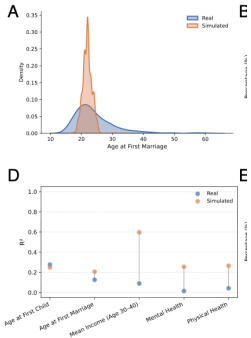
- **Cons:**
  - Poorly reflect "hard-to-reach populations" (Bollen et al., 2026); fail to capture experiential and cultural depth (Pataranutaporn et al., 2025)
  - Reflect certain partisan (Santurkar et al., 2023) or cultural biases (Tao et al., 2024)
  - Close to averages but with less variation (Bisbee et al., 2024); have strong, varying biases and have low variance in opinion or attitudinal research (Boelaert et al., 2025); likely to both misportray and flatten demographic representations (A. Wang et al., 2025); and tend to compress heterogeneity into simplified typological structures (Y. Xie & Xie, 2025)



LLM-generated answers on values deviate from those of different countries; Fig 1 in Tao et al. (2024)

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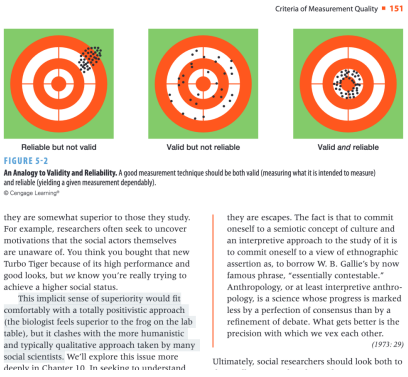
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Failing to capture the dispersion of real-world distributions; Fig 3 in Y. Xie & Xie (2025)

3.4 LLMs as Subjects: Evaluation

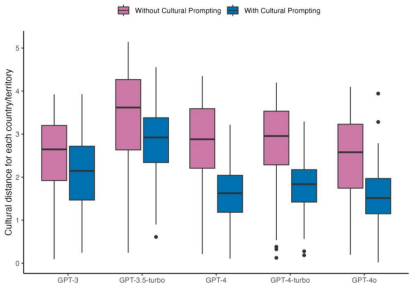
- *Validity* and *reliability* remain concerns (Zhou et al., 2026); please refer to previous lectures by Prof. Ye



Validity and reliability; from Babbie (2021)

3.5 LLMs as Subjects: Improvement

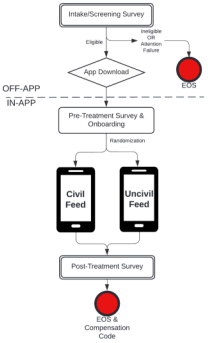
- *Validity* and *reliability* remain concerns (Zhou et al., 2026)
- Three strategies for improvement
  - Alignment (pre-use): Adjusting model outputs; mostly done by companies and in computer science studies
  - Optimization (in-use): **Fine-tuning or prompting** strategies to improve performance on specific tasks (e.g., Kolluri et al., 2025; Tao et al., 2024; Y. Xie & Xie, 2025)
  - Correction (post-use): Post-processing corrections; an emerging promise in the econometric and machine learning literature (e.g., Chernozhukov et al., 2018; Egami et al., 2024)



Prompting mitigates cultural biases; Source: Fig 2 in Tao et al. (2024)

3.6 LLMs as Methods: Assistant

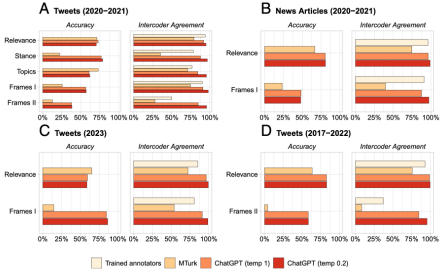
- Interviewing human participants (qualitative) (Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **Supporting experiments** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- Analysing textual content (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Synthetic communicative environment in a randomized experiment; Allamong et al. (2026)

3.6 LLMs as Methods: Assistant

- Interviewing human participants (qualitative) (Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- Supporting experiments (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- **Analysing textual content** (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- Mining social networks (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



Better annotation than crowd-workers; Gilardi et al. (2023)

TUT2

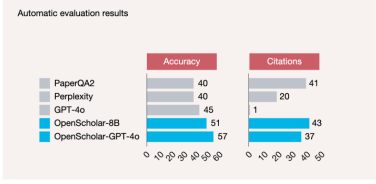
Topic classification using LLMs

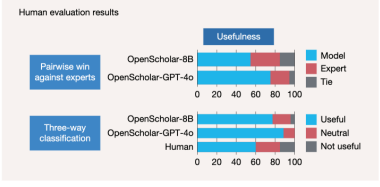
Don't forget about validation!

Online material: GitHub

3.7 LLMs as Methods: Collaborator

- **Reviewing literature** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- Proposing hypotheses (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- Designing research protocols (Baek et al., 2025; Manning et al., 2024)

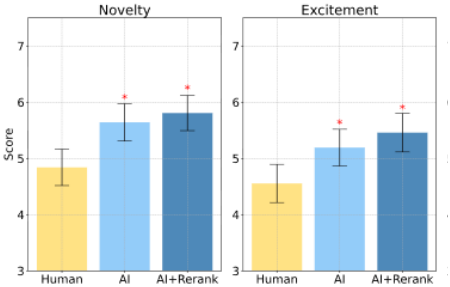




PhD-level literature review system powered by LLMs; see Asai et al. (2026)

3.7 LLMs as Methods: Collaborator

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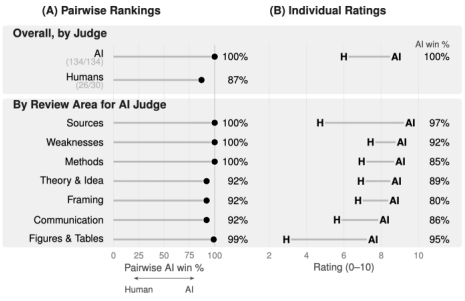
Greater novelty and excitement in LLM-generated hypotheses; see Si et al. (2024)

TUT3

Agentic Modelling in Data Analysis
Online material: GitHub

3.8 LLMs as Methods: Evaluator

- Evaluating research protocols (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **Critiquing research outcomes** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; Liang et al., 2024)



AI reviews dominate human reviews; source: (King & Worthington, 2026); not peer-reviewed but well designed

TUT4

Critically evaluate LLM-generated review comments
GitHub

3.9 LLMs as Objects

- "A communication technology that reconfigures production, circulation, interpretation, and power across media systems, raising critical questions about authenticity, access, and the reproduction of social inequalities." (Zhou et al., 2026)
- For example, Liu and Zhang (2026) evaluates how LLM-generated summaries shape public sentiment, where the treatment is the model itself.

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

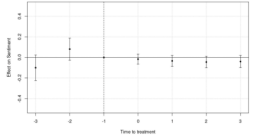


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

AIGS as objects in Liu & Zhang (2026)

Quiz

Propose one research question involving LLMs as objects in the online group chat

Please briefly clarify:

(a) the question of interest

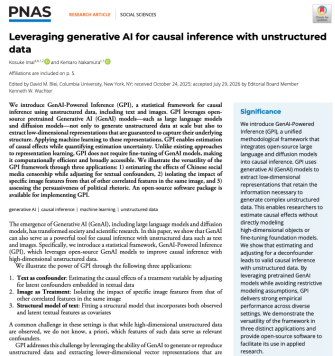
(b) operationalization of your variables

(c) the most important figure to be presented

(d) expected contribution

3.10 *LLM for inference: New frontier

- Double/debiased machine learning enables valid causal inference while flexible ML models estimate nuisance functions (Chernozhukov et al., 2017); building on this, Imai and Nakamura (2026) leverage the internal representations of generative AI to estimate causal effects of unstructured treatments such as texts and images



PNAS RESEARCH ARTICLE | SOCIAL SCIENCES

Leveraging generative AI for causal inference with unstructured data

Kosuke Imai[a,1,2] and Kentaro Nakamura[a,1]

Affiliations are included on p. 5.

Edited by David M. Blei, Columbia University, New York, NY; received October 24, 2025; accepted July 29, 2026 by Editorial Board Member Kenneth W. Wachter

We introduce GenAI-Powered Inference (GPI), a statistical framework for causal inference using unstructured data, including text and images. GPI leverages open-source pretrained Generative AI (GenAI) models—such as large language models and diffusion models—not only to generate unstructured data at scale but also to extract low-dimensional representations that are guaranteed to capture their underlying structure. Applying machine learning to these representations, GPI enables estimation of causal effects while quantifying estimation uncertainty. Unlike existing approaches to representation learning, GPI does not require fine-tuning of GenAI models, making it computationally efficient and broadly accessible. We illustrate the versatility of the GPI framework through three applications: 1) estimating the effects of Chinese social media censorship while adjusting for textual confounders, 2) isolating the impact of specific image features from that of other correlated features in the same image, and 3) assessing the persuasiveness of political rhetoric. An open-source software package is available for implementing GPI.

generative AI | causal inference | machine learning | unstructured data

The emergence of Generative AI (GenAI), including large language models and diffusion models, has transformed society and scientific research. In this paper, we show that GenAI can also serve as a powerful tool for causal inference with unstructured data such as text and images. Specifically, we introduce a statistical framework, GenAI-Powered Inference (GPI), which leverages open-source GenAI models to improve causal inference with high-dimensional unstructured data.

We illustrate the power of GPI through the following three applications:

1. **Text as confounder**: Estimating the causal effects of a treatment variable by adjusting for latent confounders embedded in textual data
2. **Image as Treatment**: Isolating the impact of specific image features from that of other correlated features in the same image
3. **Structural model of text**: Fitting a structural model that incorporates both observed and latent textual features as covariates

A common challenge in these settings is that while high-dimensional unstructured data are observed, we do not know, a priori, which features of such data serve as relevant confounders.

GPI addresses this challenge by leveraging the ability of GenAI to generate or reproduce unstructured data and extracting lower-dimensional vector representations that are guaranteed to contain all the relevant information. Machine learning is then applied to these representations to estimate a *deconfounder*, which is a summary of confounding

Significance

We introduce GenAI-Powered Inference (GPI), a unified methodological framework that integrates open-source large language and diffusion models into causal inference. GPI uses generative AI (GenAI) models to extract low-dimensional representations that retain the information necessary to generate complex unstructured data. This enables researchers to estimate causal effects without directly modeling high-dimensional objects or fine-tuning foundation models. We show that estimating and adjusting for a deconfounder leads to valid causal inference with unstructured data. By leveraging pretrained GenAI models while avoiding restrictive modeling assumptions, GPI delivers strong empirical performance across diverse settings. We demonstrate the versatility of the framework in three distinct applications and provide open-source software to facilitate its use in applied research.

Source: Imai & Nakamura (2026)

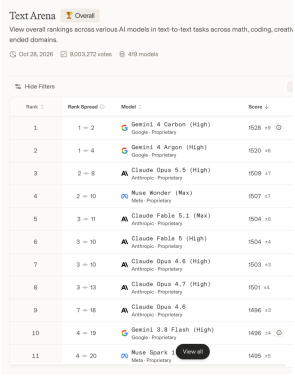
Deploy Responsively

4.1 LLMs deployed responsibly: Selection

- Consider a model as a function, $f(data, algo, object)$; differences in each part substantially shape its in-domain capabilities (like better coding or writing)
- In this sense, we may need to choose different models for different tasks, depending on their strengths and limitations

4.1 LLMs deployed responsibly: Selection

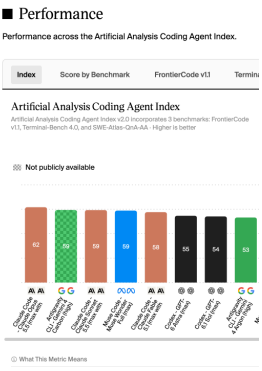
- For example, you may expect Gemini Carbon to excel at writing tasks, whereas Claude Opus 5.5 outperforms in engineering



Source: https://arena.ai/leaderboard/chat/text (Nov 3, 2026)

4.1 LLMs deployed responsibly: Selection

- For example, you may expect Gemini Carbon to excel at writing tasks, whereas Claude Opus 5.5 outperforms in engineering
- In general, model labs follow different commercialization strategies; a lab whose business model targets engineering tasks may prioritize coding capabilities (e.g., Anthropic); please check benchmarks / review the literature / do a pilot test before using a model



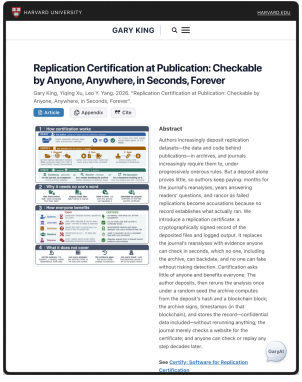
Source: https://artificialanalysis.ai/agents/coding-agents#coding-agents-index (Nov 3, 2026)

4.2 LLMs deployed responsibly: Validity and Reliability

- Again, always verify the validity and reliability of the model outputs before using them in your research

4.3 LLMs deployed responsibly: Reproducibility

- Reproducibility is important to science
- Ensure that the AI part of your research is reproducible by:
  - Preferring API-based access over chat-based access
  - Documenting model versions, prompts, thinking level, and parameters
  - Providing detailed procedures
  - Sharing code and data
  - Fixing the random seed and temperature (search for details)



Replication certification at publication; Source: King et al. (2026)

*TUT5 & 6

Multimodal object detection and controlled stimulus generation
Online material: GitHub

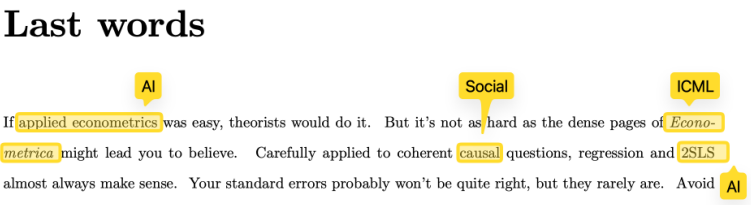
Takeaways

5.1 Takeaways

- AI models (including LLMs) are just tools for high-dimensional data analysis; they can be subjects, objects, and methods in quantitative social research
- As methods, they are expected to be used in a responsible way in which validity, reliability, and reproducibility are carefully considered
- Like any other methods, although they evolve rapidly, their use should rest on our taste for good questions in research, industry, and life, which remains central to higher education.

5.1 Takeaways

- "... Be skeptic... don't panic!" (Angrist & Pischke, 2009)



Source: Angrist & Pischke (2009)

Resources

6.1 Resources

- **Locate the most appropriate AI model for your task**
  - Artificial Analysis https://artificialanalysis.ai/
  - Arena.ai, blind benchmarking for AI models https://arena.ai/leaderboard/
- **Find problems in your manuscript**
  - Refine, publication-level technical review; adopted by AEA https://www.refine.ink/
  - OpenAIReview, open-source multi-agent manuscript review solution by UChicago HAI https://github.com/ChicagoHAI/OpenAIReview
- **Double-check references and citations**
  - CiteVerify, "a local tool for verifying references in research papers", https://github.com/cuihuash/CiteVerify
  - Chatting with frontier models also helps

6.1 Resources

- **Organize multiple AI models and services**
  - For macOS, magpie, a local AI router manager for Mac https://github.com/yetone/magpie
  - For other platforms, cc-switch, https://github.com/farion1231/cc-switch
- **Read scientific literature in your own language**
  - PDFMathTranslate, read scientific PDFs in your language https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)

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References

7.1 References I

Allamong, M. B., Bail, C., Hillygus, D. S., & Volfovsky, A. (2026). The Causal Effects of Political Incivility in Social Media Discussions. Retrieved March 17, 2026, from https://osf.io/download/yaq29

Angrist, J. D., & Pischke, J.-S. (2009). *Mostly harmless econometrics: An empiricist's companion*. Princeton university press.

Argyle, L. P., Bail, C. A., Busby, E. C., Gubler, J. R., Howe, T., Rytting, C., Sorensen, T., & Wingate, D. (2023). Leveraging AI for democratic discourse: Chat interventions can improve online political conversations at scale. *Proceedings of the National Academy of Sciences of the United States of America*, *120*(41), e2311627120. https://doi.org/10.1073/pnas.2311627120

Argyle, L. P., Busby, E. C., Fulda, N., Gubler, J. R., Rytting, C., & Wingate, D. (2023). Out of One, Many: Using Language Models to Simulate Human Samples. *Political Analysis*, *31*(3), 337–351. https://doi.org/10.1017/pan.2023.2

Asai, A., He, J., Shao, R., Shi, W., Singh, A., Chang, J. C., Lo, K., Soldaini, L., Feldman, S., D'Arcy, M., Wadden, D., Latzke, M., Sparks, J., Hwang, J. D., Kishore, V., Tian, M., Ji, P., Liu, S., Tong, H., … Hajishirzi, H. (2026). Synthesizing scientific literature with retrieval-augmented language models. *Nature*, *650*(8103), 857–863. https://doi.org/10.1038/s41586-025-10072-4

Ashery, A. F., Aiello, L. M., & Baronchelli, A. (2025). Emergent social conventions and collective bias in LLM populations. *Science Advances*, *11*(20), eadu9368. https://doi.org/10.1126/sciadv.adu9368

Babbie, E. R. (2021). *The practice of social research* (15th ed.). Cengage.

Baek, J., Jauhar, S. K., Cucerzan, S., & Hwang, S. J. (2025). ResearchAgent: Iterative Research Idea Generation over Scientific Literature with Large Language Models. In L. Chiruzzo, A. Ritter, & L. Wang (Eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)* (pp. 6709–6738). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.naacl-long.342

Bisbee, J., Clinton, J. D., Dorff, C., Kenkel, B., & Larson, J. M. (2024). Synthetic Replacements for Human Survey Data? The Perils of Large Language Models. *Political Analysis*, *32*(4), 401–416. https://doi.org/10.1017/pan.2024.5

Boelaert, J., Coavoux, S., Ollion, É., Petev, I., & Präg, P. (2025). Machine Bias. How Do Generative Language Models Answer Opinion Polls?1. *Sociological Methods & Research*, *54*(3), 1156–1196. https://doi.org/10.1177/00491241251330582

Bollen, P., Higton, J., & Sands, M. (2026). Nationally Representative, Locally Misaligned: The Biases of Generative Artificial Intelligence in Neighborhood Perception. *Political Analysis*, *34*(3), 479–487. https://doi.org/10.1017/pan.2025.10022

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., & Newey, W. (2017). Double/Debiased/Neyman Machine Learning of Treatment Effects. *American Economic Review*, *107*(5), 261–265. https://doi.org/10.1257/aer.p20171038

7.1 References II

Chernozhukov, V., Chetverikov, D., Demirer, M., Duflo, E., Hansen, C., Newey, W., & Robins, J. (2018). Double/debiased machine learning for treatment and structural parameters. *The Econometrics Journal*, *21*(1), C1–C68. https://doi.org/10.1111/ectj.12097

Chew, R., Bollenbacher, J., Wenger, M., Speer, J., & Kim, A. (2023, June 23). *LLM-Assisted Content Analysis: Using Large Language Models to Support Deductive Coding*. arXiv: 2306.14924 [cs.CL]. https://doi.org/10.48550/arXiv.2306.14924

Dennstädt, F., Zink, J., Putora, P. M., Hastings, J., & Cihoric, N. (2024). Title and abstract screening for literature reviews using large language models: An exploratory study in the biomedical domain. *Systematic Reviews*, *13*(1), 158. https://doi.org/10.1186/s13643-024-02575-4

Dillion, D., Tandon, N., Gu, Y., & Gray, K. (2023). Can AI language models replace human participants? *Trends in Cognitive Sciences*, *27*(7), 597–600. https://doi.org/10.1016/j.tics.2023.04.008

Dominguez-Olmedo, R., Hardt, M., & Mendler-Dünner, C. (2024, December 9). *Questioning the Survey Responses of Large Language Models*. arXiv: 2306.07951 [cs.CL]. https://doi.org/10.48550/arXiv.2306.07951

Egami, N., Hinck, M., Stewart, B. M., & Wei, H. (2024). Using large language model annotations for the social sciences: A general framework of using predicted variables in downstream analyses. *Preprint from November*, *17*, 2024. Retrieved May 12, 2026, from https://naokiegami.com/paper/dsl_ss.pdf

Feng, S., Wan, H., Wang, N., Tan, Z., Luo, M., & Tsvetkov, Y. (2024). What Does the Bot Say? Opportunities and Risks of Large Language Models in Social Media Bot Detection. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 3580–3601). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.196

Gao, C., Lan, X., Lu, Z., Mao, J., Piao, J., Wang, H., Jin, D., & Li, Y. (2025, June 3). *S^3: Social-network Simulation System with Large Language Model-Empowered Agents*. arXiv: 2307.14984 [cs.SI]. https://doi.org/10.48550/arXiv.2307.14984

Gehrmann, J., Quakulinski, L., & Beyan, O. (2024). Large Language Models for Literature Reviews - An Exemplary Comparison of LLM-based Approaches with Manual Methods. *2024 2nd International Conference on Foundation and Large Language Models (FLLM)*, 385–391. https://doi.org/10.1109/FLLM63129.2024.10852447

Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd-workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, *120*(30), e2305016120. https://doi.org/10.1073/pnas.2305016120

Goyal, T., Li, J. J., & Durrett, G. (2023, May 23). *News Summarization and Evaluation in the Era of GPT-3*. arXiv: 2209.12356 [cs.CL]. https://doi.org/10.48550/arXiv.2209.12356

7.1 References III

Grimmer, J., Roberts, M. E., & Stewart, B. M. (2021). Machine Learning for Social Science: An Agnostic Approach. *Annual Review of Political Science*, *24*, 395–419. https://doi.org/10.1146/annurev-polisci-053119-015921

Hackenburg, K., & Margetts, H. (2024). Evaluating the persuasive influence of political microtargeting with large language models. *Proceedings of the National Academy of Sciences of the United States of America*, *121*(24), e2403116121. https://doi.org/10.1073/pnas.2403116121

Hai Liang. (2026). *A Computational Approach to Human - AI Communication*. Singapore. www.dropbox.com/scl/fi/nlq3k73akoln87ebkkxsr/NUS.pdf?rlkey=0kzml5htrvthlhvqa6m6w9ee2&e=1&dl=0

Hämäläinen, P., Tavast, M., & Kunnari, A. (2023). Evaluating Large Language Models in Generating Synthetic HCI Research Data: A Case Study. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 1–19. https://doi.org/10.1145/3544548.3580688

Hanschmann, L., Mokelke, M., & Maedche, A. (2025). LadderChat An LLM-Based Conversational Agent for Laddering Interviews. In A. Følstad, S. Papadopoulos, T. Araujo, E. L.-C. Law, E. Luger, S. Hobert, & P. B. Brandtzaeg (Eds.), *Chatbots and Human-Centered AI* (pp. 48–65). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-88045-2_4

Imai, K., & Nakamura, K. (2026). Leveraging generative AI for causal inference with unstructured data. *Proceedings of the National Academy of Sciences*, *123*(36), e2530532123. https://doi.org/10.1073/pnas.2530532123

Jiang, Y., Qiu, R., Zhang, Y., & Zhang, P.-F. (2024). Balanced and Explainable Social Media Analysis for Public Health with Large Language Models. In Z. Bao, R. Borovica-Gajic, R. Qiu, F. Choudhury, & Z. Yang (Eds.), *Databases Theory and Applications* (pp. 73–86). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-47843-7_6

Ke, P., Wen, B., Feng, A., Liu, X., Lei, X., Cheng, J., Wang, S., Zeng, A., Dong, Y., Wang, H., Tang, J., & Huang, M. (2024). CritiqueLLM: Towards an Informative Critique Generation Model for Evaluation of Large Language Model Generation. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 13034–13054). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.704

King, G., & Worthington, S. (2026). AI Article Reviews Dominate Human Peer Reviews at Their Best: Evidence and Policy Choices. Retrieved November 4, 2026, from https://gking.harvard.edu/peerless/

King, G., Xu, Y., & Yang, L. Y. (2026). *Replication certification at publication: Checkable by anyone, anywhere, in seconds, forever.*

Kolluri, A., Wu, S., Park, J. S., & Bernstein, M. S. (2025). Finetuning LLMs for Human Behavior Prediction in Social Science Experiments. In C. Christodoulopoulos, T. Chakraborty, C. Rose, & V. Peng (Eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing* (pp. 30096–30111). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-main.1530

7.1 References IV

Li, H., Yu, J., Li, R., Hao, Z., Xuan, Y., Yuan, J., Xu, B., Li, J., & Liu, Z. (2024). LM-Interview: An Easy-to-use Smart Interviewer System via Knowledge-guided Language Model Exploitation. In D. I. Hernandez Farias, T. Hope, & M. Li (Eds.), *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 520–528). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.emnlp-demo.52

Li, M., Lau, J. H., & Hovy, E. (2024). A Sentiment Consolidation Framework for Meta-Review Generation. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 10158–10177). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.547

Liang, W., Zhang, Y., Cao, H., Wang, B., Ding, D. Y., Yang, X., Vodrahalli, K., He, S., Smith, D. S., Yin, Y., McFarland, D. A., & Zou, J. (2024). Can Large Language Models Provide Useful Feedback on Research Papers? A Large-Scale Empirical Analysis. *NEJM AI*, *1*(8), AIoa2400196. https://doi.org/10.1056/AIoa2400196

Lin, Y.-T., & Chen, Y.-N. (2023). LLM-Eval: Unified Multi-Dimensional Automatic Evaluation for Open-Domain Conversations with Large Language Models. In Y.-N. Chen & A. Rastogi (Eds.), *Proceedings of the 5th Workshop on NLP for Conversational AI (NLP4ConvAI 2023)* (pp. 47–58). Association for Computational Linguistics. https://doi.org/10.18653/v1/2023.nlp4convai-1.5

Liu, A. Y., & Zhang, M. M. (2026). Positivity Bias in AI-Generated Summaries of User-Generated Content: Exploring Its Sources and Impact on Public Sentiment. *Computational Communication Research*, *8*(2), 1. https://doi.org/10.5117/CCR2026.2.4.LIU

Luceri, L., Boniardi, E., & Ferrara, E. (2024). Leveraging Large Language Models to Detect Influence Campaigns on Social Media. *Companion Proceedings of the ACM Web Conference 2024*, 1459–1467. https://doi.org/10.1145/3589335.3651912

Manning, B. S., Zhu, K., & Horton, J. J. (2024, April). *Automated Social Science: Language Models as Scientist and Subjects*. 32381. https://doi.org/10.3386/w32381

Ouyang, R., Chu, C., Xin, Z., & Ma, X. (2025). PDFMathTranslate: Scientific document translation preserving layouts. In I. Habernal, P. Schulam, & J. Tiedemann (Eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: System Demonstrations* (pp. 918–924). Association for Computational Linguistics. https://doi.org/10.18653/v1/2025.emnlp-demos.71

Park, J. S., O'Brien, J., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative Agents: Interactive Simulacra of Human Behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 1–22. https://doi.org/10.1145/3586183.3606763

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023, August 6). *Generative Agents: Interactive Simulacra of Human Behavior*. arXiv: 2304.03442 [cs.HC]. https://doi.org/10.48550/arXiv.2304.03442

7.1 References V

Park, J. S., Zou, C. Q., Kamphorst, J., Egan, N., Shaw, A., Hill, B. M., Cai, C., Morris, M. R., Liang, P., Willer, R., & Bernstein, M. S. (2026, June 28). *LLM Agents Grounded in Self-Reports Enable General-Purpose Simulation of Individuals*. arXiv: 2411.10109 [cs.AI]. https://doi.org/10.48550/arXiv.2411.10109

Pataranutaporn, P., Powdthavee, N., Archiwaranguprok, C., & Maes, P. (2025). Simulating human well-being with large language models: Systematic validation and misestimation across 64,000 individuals from 64 countries. *Proceedings of the National Academy of Sciences*, *122*(48), e2519394122. https://doi.org/10.1073/pnas.2519394122

Poole, D. L., & Mackworth, A. K. (2023, July 13). *Artificial Intelligence: Foundations of Computational Agents*. Cambridge University Press. https://doi.org/10.1017/9781009258227

Santurkar, S., Durmus, E., Ladhak, F., Lee, C., Liang, P., & Hashimoto, T. (2023). Whose Opinions Do Language Models Reflect? *Proceedings of the 40th International Conference on Machine Learning*, 29971–30004. Retrieved November 3, 2026, from https://proceedings.mlr.press/v202/santurkar23a.html

Shankar, S., Zamfirescu-Pereira, J., Hartmann, B., Parameswaran, A., & Arawjo, I. (2024). Who Validates the Validators? Aligning LLM-Assisted Evaluation of LLM Outputs with Human Preferences. *Proceedings of the 37th Annual ACM Symposium on User Interface Software and Technology*, 1–14. https://doi.org/10.1145/3654777.3676450

Si, C., Yang, D., & Hashimoto, T. (2024, September 6). *Can LLMs Generate Novel Research Ideas? A Large-Scale Human Study with 100+ NLP Researchers*. arXiv: 2409.04109 [cs.CL]. https://doi.org/10.48550/arXiv.2409.04109

Tao, Y., Viberg, O., Baker, R. S., & Kizilcec, R. F. (2024). Cultural bias and cultural alignment of large language models (M. Muthukrishna, Ed.). *PNAS Nexus*, *3*(9), pgae346. https://doi.org/10.1093/pnasnexus/pgae346

Tuggener, D., Schneider, T., Huwiler, A., Kreienbühl, T., Hischier, S., von Däniken, P., & Niehaus, S. (2024). Role-Playing LLMs in Professional Communication Training: The Case of Investigative Interviews with Children. In P. H. Luz de Araujo, A. Baumann, D. Gromann, B. Krenn, B. Roth, & M. Wiegand (Eds.), *Proceedings of the 20th Conference on Natural Language Processing (KONVENS 2024)* (pp. 249–263). Association for Computational Linguistics. Retrieved November 4, 2026, from https://aclanthology.org/2024.konvens-main.26/

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, *30*. https://papers.nips.cc/paper/7181-attention-is-all-you-need

Velez, Y. R., & Liu, P. (2025). Confronting Core Issues: A Critical Assessment of Attitude Polarization Using Tailored Experiments. *American Political Science Review*, *119*(2), 1036–1053. https://doi.org/10.1017/S0003055424000819

7.1 References VI

Wang, A., Morgenstern, J., & Dickerson, J. P. (2025). Large language models that replace human participants can harmfully misportray and flatten identity groups. *Nature Machine Intelligence*, *7*(3), 400–411. https://doi.org/10.1038/s42256-025-00986-z

Wang, L., Zhang, J., Yang, H., Chen, Z.-Y., Tang, J., Zhang, Z., Chen, X., Lin, Y., Sun, H., Song, R., Zhao, X., Xu, J., Dou, Z., Wang, J., & Wen, J.-R. (2025). User Behavior Simulation with Large Language Model-based Agents. *ACM Transactions on Information Systems*, *43*(2), 55:1–55:37. https://doi.org/10.1145/3708985

Wang, Q., Downey, D., Ji, H., & Hope, T. (2024). SciMON: Scientific Inspiration Machines Optimized for Novelty. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 279–299). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.18

Wang, X., Kim, H., Rahman, S., Mitra, K., & Miao, Z. (2024). Human-LLM Collaborative Annotation Through Effective Verification of LLM Labels. *Proceedings of the 2024 CHI Conference on Human Factors in Computing Systems*, 1–21. https://doi.org/10.1145/3613904.3641960

Xie, C., Chen, C., Jia, F., Ye, Z., Lai, S., Shu, K., Gu, J., Bibi, A., Hu, Z., Jurgens, D., Evans, J., Torr, P., Ghanem, B., & Li, G. (2024, November 1). *Can Large Language Model Agents Simulate Human Trust Behavior?* arXiv: 2402.04559 [cs.AI]. https://doi.org/10.48550/arXiv.2402.04559

Xie, Y., & Xie, Y. (2025, March 2). *Variance reduction in output from generative AI*. arXiv: 2503.01033 [cs.CY]. https://doi.org/10.48550/arXiv.2503.01033

Xu, F., Shao, C., Chen, L., Zeng, Q., Chen, Z., Sukiennik, N., Wang, J., Gao, C., Wang, H., Lian, J., Meng, Q., Xie, X., Li, Y., & Evans, J. (2026). Large language models for computational social science. *Nature Human Behaviour*. https://doi.org/10.1038/s41562-026-02539-9

Yan, Y., Zeng, Q., Zheng, Z., Yuan, J., Feng, J., Zhang, J., Xu, F., & Li, Y. (2024, October 11). *OpenCity: A Scalable Platform to Simulate Urban Activities with Massive LLM Agents*. arXiv: 2410.21286 [cs.MA]. https://doi.org/10.48550/arXiv.2410.21286

Yang, Z., Du, X., Li, J., Zheng, J., Poria, S., & Cambria, E. (2024). Large Language Models for Automated Open-domain Scientific Hypotheses Discovery. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Findings of the Association for Computational Linguistics: ACL 2024* (pp. 13545–13565). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.findings-acl.804

Yu, J., Ren, Y., Gong, C., Tan, J., Li, X., & Zhang, X. (2025). Leveraging Large Language Models for Node Generation in Few-Shot Learning on Text-Attributed Graphs. *Proceedings of the AAAI Conference on Artificial Intelligence*, *39*(12), 13087–13095. https://doi.org/10.1609/aaai.v39i12.33428

7.1 References VII

Zhang, J., Xu, X., Zhang, N., Liu, R., Hooi, B., & Deng, S. (2024). Exploring Collaboration Mechanisms for LLM Agents: A Social Psychology View. In L.-W. Ku, A. Martins, & V. Srikumar (Eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)* (pp. 14544–14607). Association for Computational Linguistics. https://doi.org/10.18653/v1/2024.acl-long.782

Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E. P., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023, December 24). *Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena*. arXiv: 2306.05685 [cs.CL]. https://doi.org/10.48550/arXiv.2306.05685

Zhou, A., Mukerjee, S., Maslowska, E., Duan, Z., Jaidka, K., Lukito, J., Shugars, S., Smith, M. A., Trilling, D., & Yang, T. (2026). Be Careful What You Prompt For: Generative AI in Computational Communication Research. *Computational Communication Research*, *8*(2), 1–34. https://doi.org/10.5117/CCR2026.2.1.ZHOU